UNIVERSITY OF HAWAIʻI

MĀLAMALAMA

1907

UA MAU KE EA O KA ʻĀINA I KA PONO

Enhancing
Segmented

B. C. Crow[1], M. A. A. Dornfest[1]
[1] Departmen

Neutrino 2026 [Un

ng Angular Sensi

d Antineutrino D

for Reactor Monitoring

est[1], J. G. Learned[1], V. A. Li[2], T. Peters[1], J. D. Seligman[1]

ment of Physics and Astronomy, University of Hawai'i at M

[2] Lawrence Livermore National Laboratory

[University of California at Irvine / June 22-

sitivity of
Detectors

an[1], N. S. Sibert[1], J. G. Yepez[1]
t Mānoa

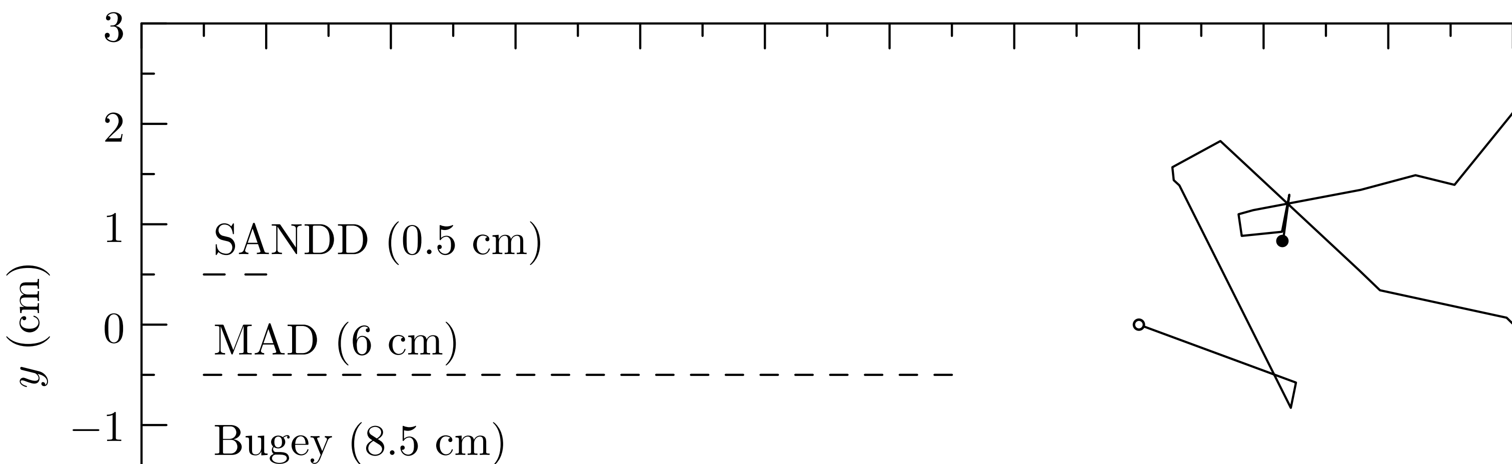
22-26, 2026]

Lawrence Livermore
National Laboratory

Consortium for Monitoring Technology and Verification

Localization of IBD Source

- Directional method for segmented Inverse Beta Dec
- Positron (e^+) gets most of the energy, and annihilat
- Neutron (^1n) gets momentum direction, and captur
- Lithium 6 dopant localizes ^1n capture with $^3_1$H and $^4_2$
  $\sim 10\ \mu$m of capture
- direction to source inferred from ^1n direction with e



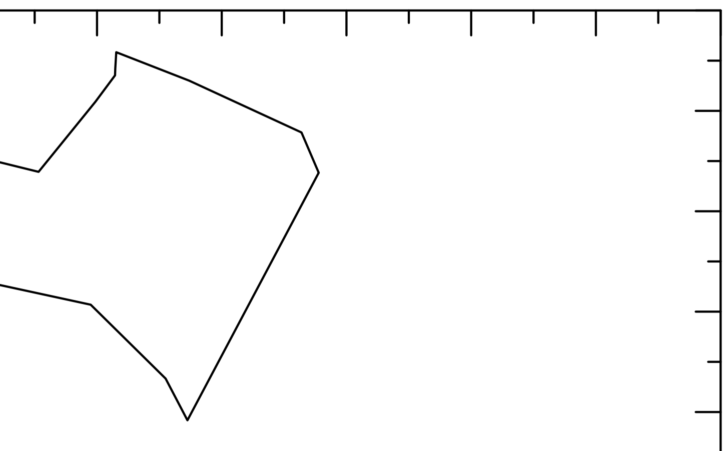
Decay (IBD) detectors

ilation marks prompt vertex

ture marks delayed vertex

nd $^4_2$He depositing energy within

th enough events



Our Approa

- Used RATPAC-2
- Created simple mo
- Created grid overl segment sizes in a

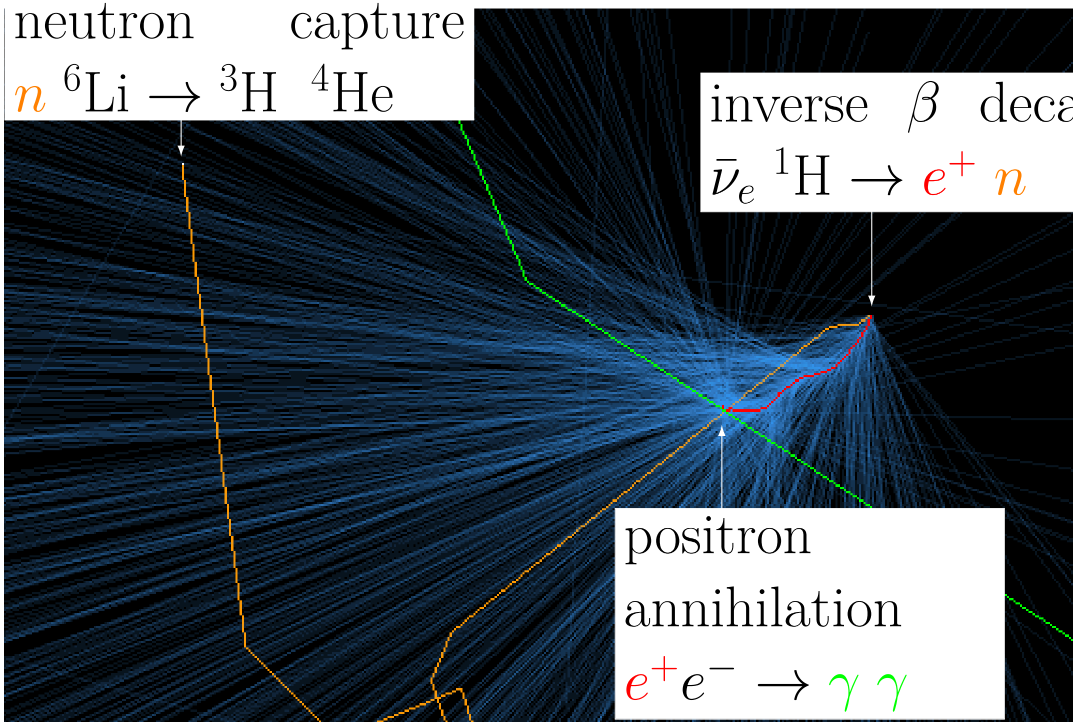
...oach to Simulation Design

...-2 for all simulations

... monolithic geometry using 0.1%-wt ^{6}Li-doped plastic scint...

...verlay to create binning matrices for 5 mm, 50 mm and 15...

...n analysis work



cintillator

150 mm

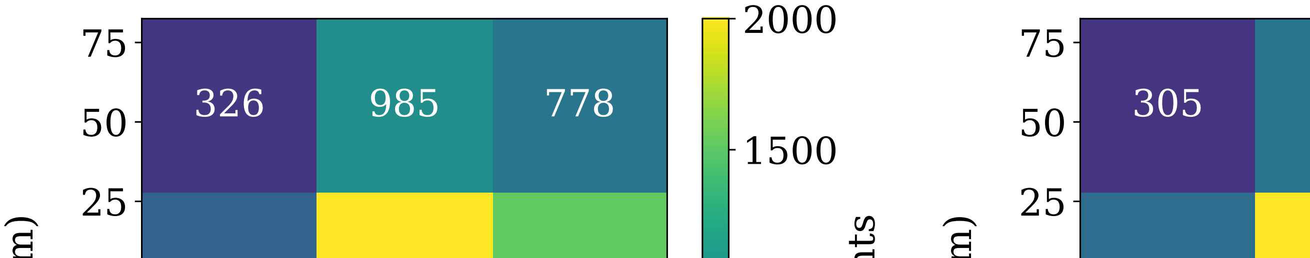
★ Conclusions, Findings,

Key findings:

- At large n, the algorithm gives the direct
- At small n, the algorithm gives a reasona
- As a pattern matching method, the algori

Future work:

- Developing methods to take advantage o
- Investigating how to apply to monolithic
- Investigating a generalized approach to t



gs, and Future Work

rection of antineutrino sources

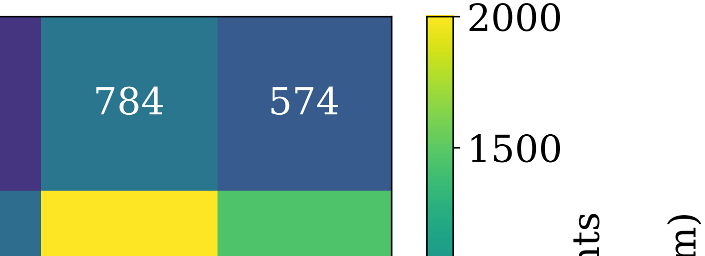
sonable uncertainty

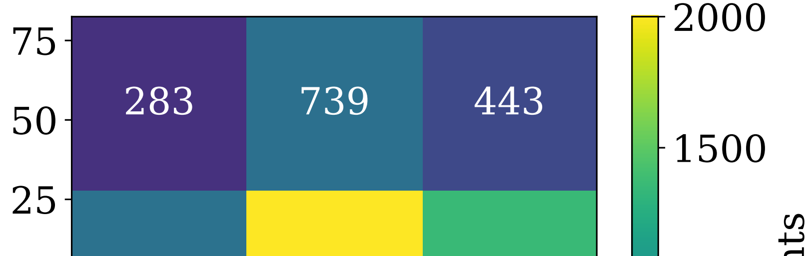
gorithm provides a comparison between distributions

ge of machine learning

thic detectors

to three dimensionally segmented detectors





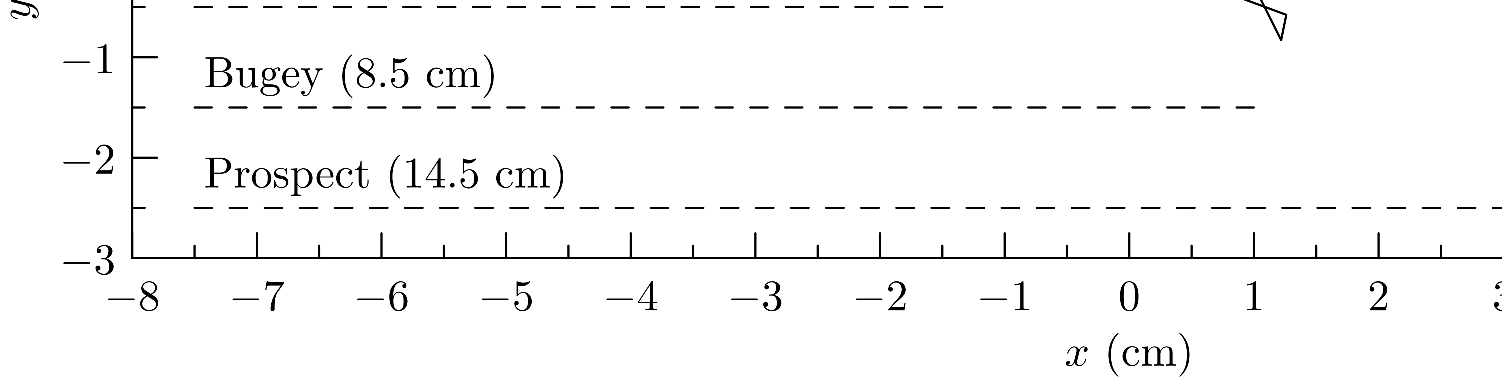
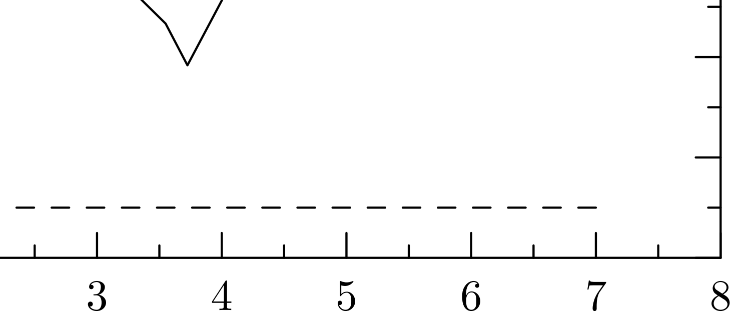


Figure 1: Diagram showcasing relative scale for characteristic neut
IBD vertex to capture location, as well as dimensions of segments
SANDD, MAD, Bugey, and PROSPECT.

Traditional Approach

- Conventional uncertainty calculation:

$$\Delta \ldots = \arctan\left(\frac{P/l}{\ldots}\right)$$



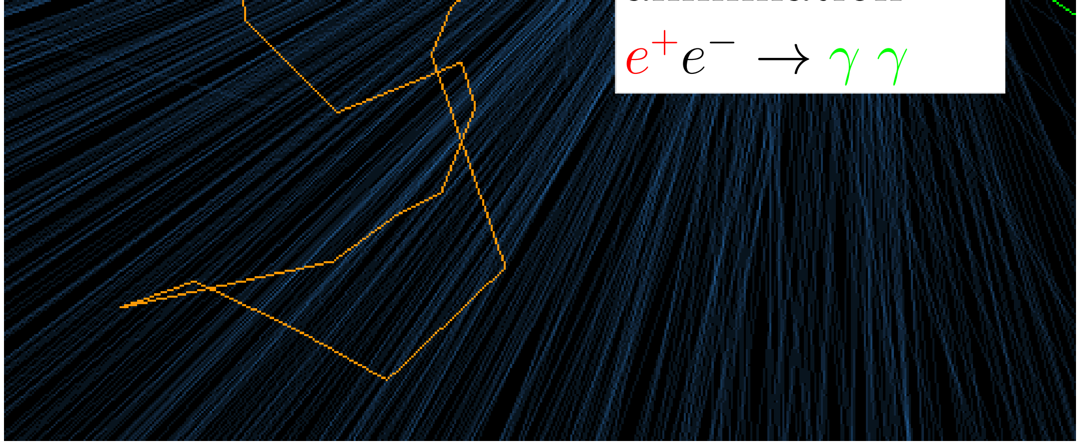
neutron xy-trajectory projection from
ents in four 2D-segmented detectors:

(1)

Figure 3: Rendering of RA
gamma-rays — green, Ch
Cherenkov photons indica
bottom left in this perspe

Novel Algor

- Generate reference
  This is done by rot
  the binning descri



f RAT-PAC 2 IBD visualization. Positron track in red, neutron track —
Cherenkov photons — cyan. Scintillation light turned off for clarity.
dicate the direction of the positron, which traveled from the top right t
rspective (neutrino, not shown, is from top right to bottom left.

orithm for Segmented Detectors

ence data for all incident angles
rotating the grid overlay on simulation data and then perfo
scribed below

t — orange,
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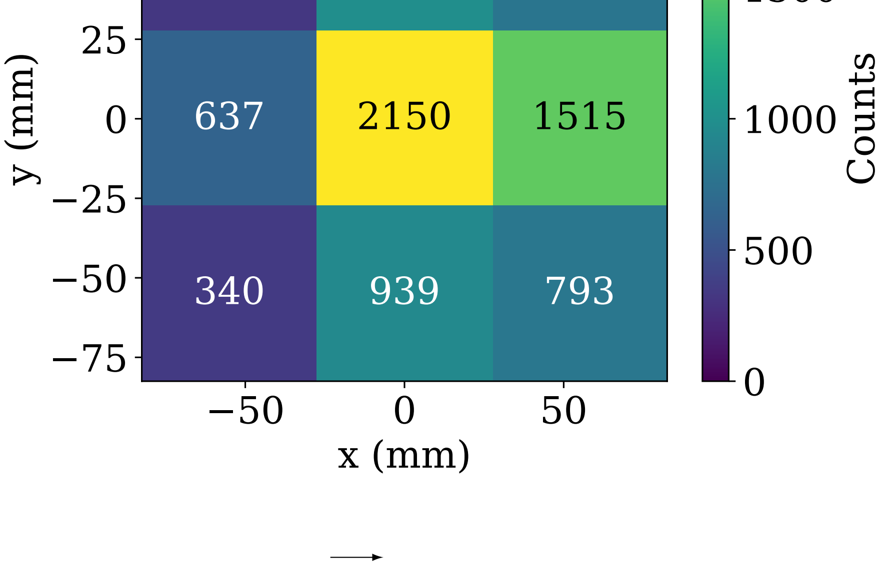
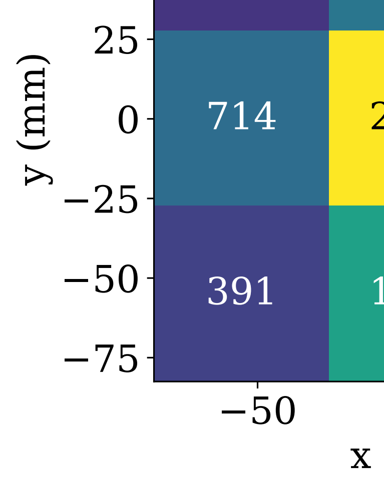
ght to the

s

erforming

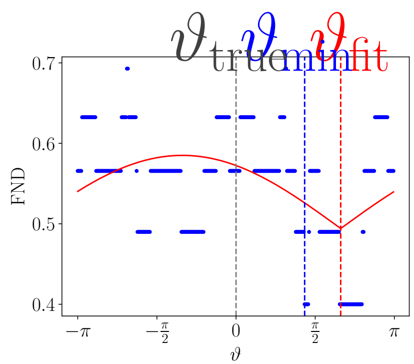


$\vartheta_\nu = 0^\circ$

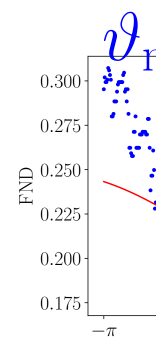


ϑ_ν

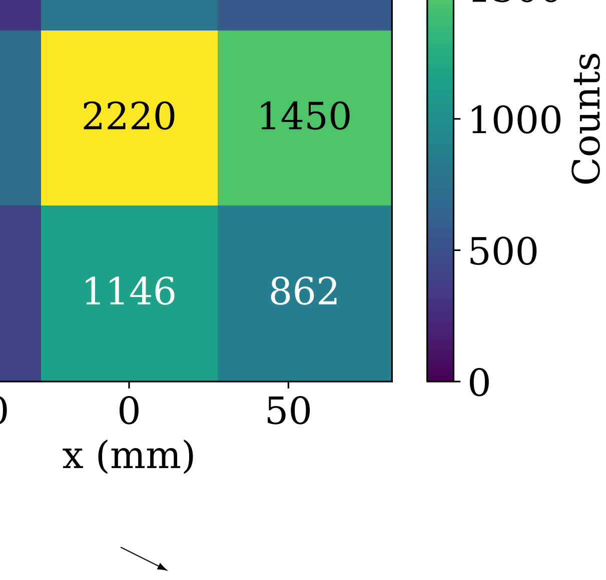
Figure 5: Examples of binning distributions for inco
direction of antineutrinos is indicated by the arrows



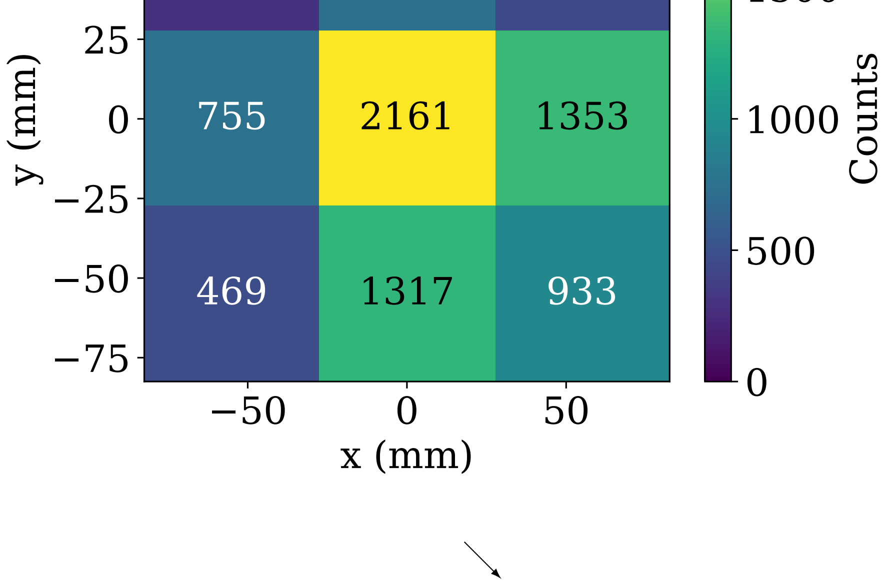
(a) $n = 10$



(b)

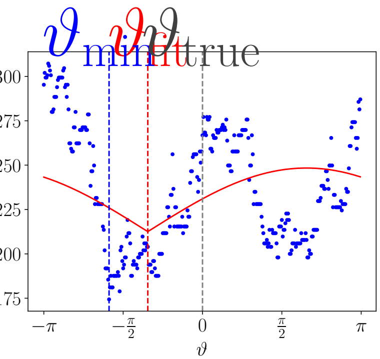


$\vartheta_\nu = -30°$

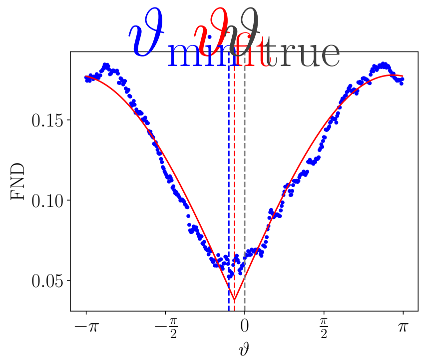


$\vartheta_\nu = -45°$

ncoming angles $\vartheta_\nu = 0°, -30°, -45°$, 0.1% ^{6}Li-doped. The
ows below the plots.



b) $n = 100$



(c) $n = 1000$

$$\Delta\varphi = \arctan\left(\frac{P/l}{\sqrt{N}}\right)$$

- If mean position resolution $P \to 0$, angular uncerta
- Vogel and Beacom model shows a large spread ($\sim$ 26
- Neutrons scatter many times before capture

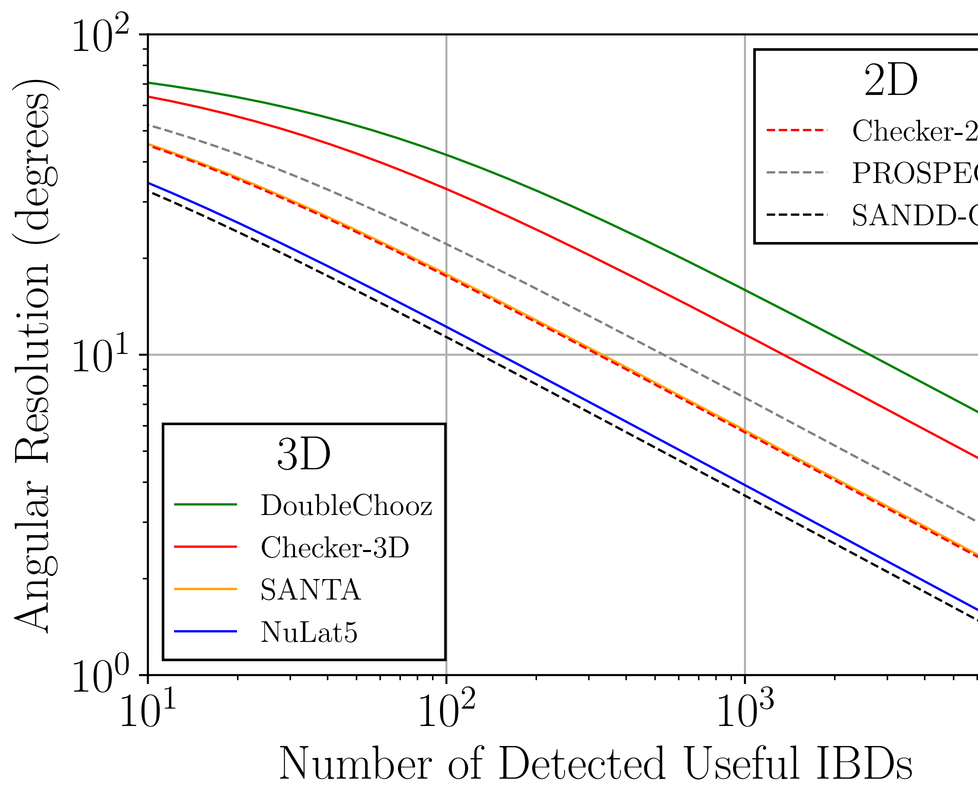
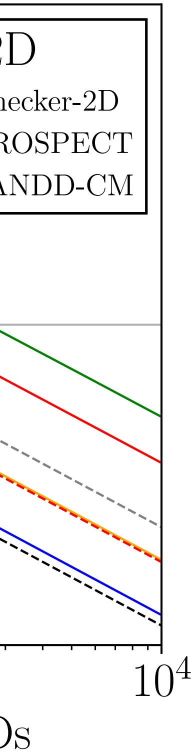


Figure 2: Angular uncertainty calculated using the conventional "

$$\left.\right) \tag{1}$$

ertainty $\Delta\varphi \to 0$ – unphysical!

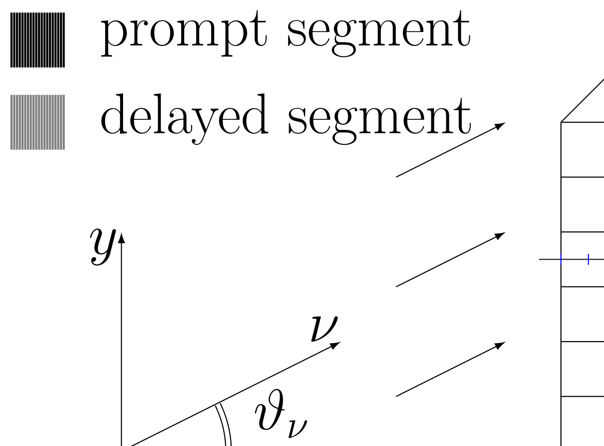
$\sim 26°$) in intial neutron direction



al “Chooz" approach. A selected

the binning descri

- Bin prompt and c number of voxels s all matrices for ref
- Create binning ma
- Compute the L^2 n
- Fit $\alpha \left|\sin\left(\frac{\theta-\theta_0}{2}\right)\right|$ to parameter for the



scribed below

d delayed displacement vectors in matrix form by countin

els separating the IBD vertex and the capture vertex. Nor

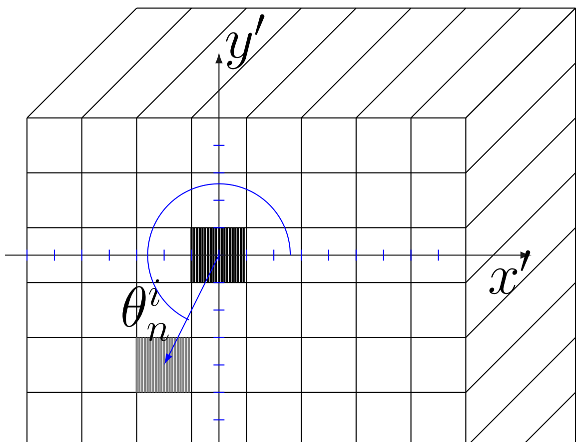
reference angles

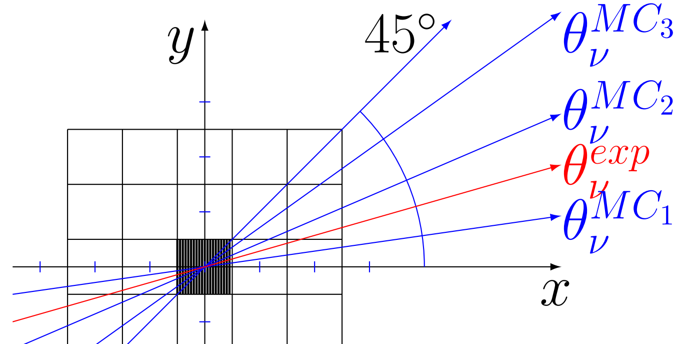
matrix for data set and normalize it

L^2 norm for the matrix differences for all reference angles

to L^2 norms as a function of angle. The minimum gives the

he direction of the neutrino source.





nting the
Normalize

s

the θ_0 fit

C_3

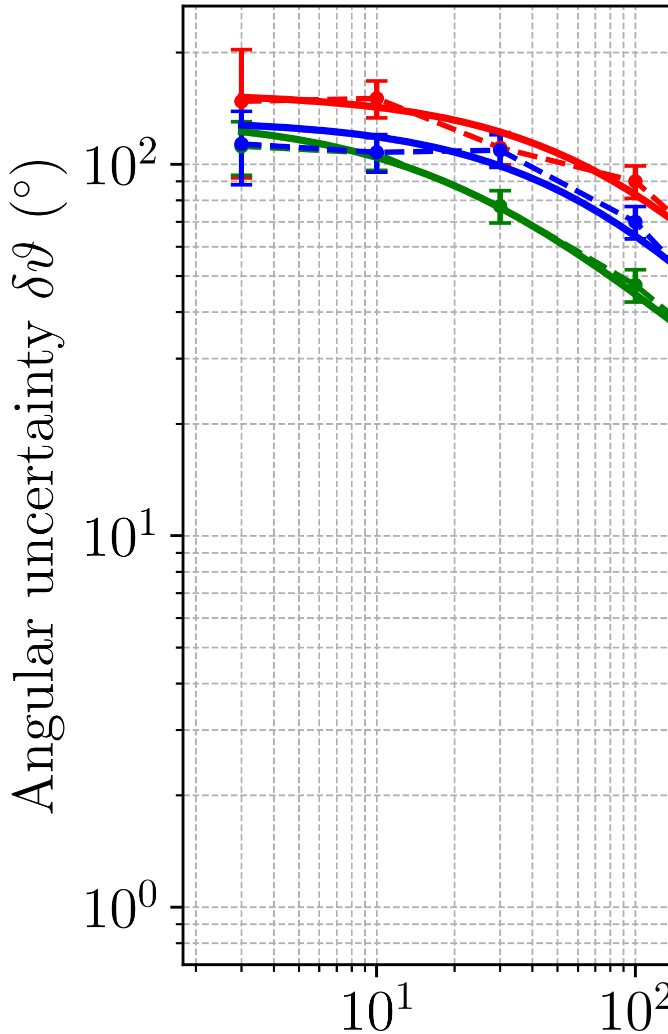
C_2

C_1

(a) $n = 10$

(b)

Figure 6: Explanation of angles and spread. The FN
data with a segment size of 50 mm.



b) $n = 100$ (c) $n = 1000$

FND here is calculated from the RATPAC2 neutron capture

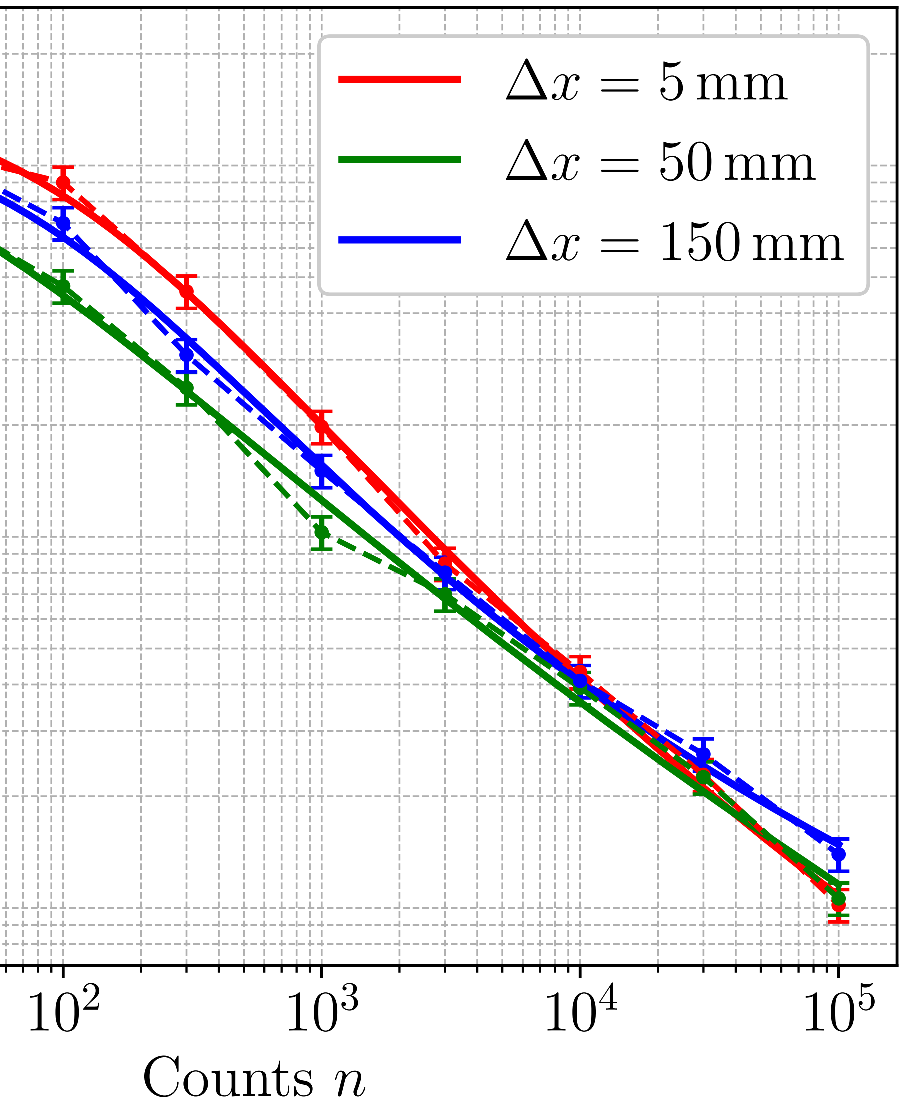


Figure 2: Angular uncertainty calculated using the conventional "
number of segmented detectors along with Chooz is highlighted.
have prompt and delayed signals in separate segments. This resul
study [**?**].

References

[1] Crow *et al.*, in prep. (2026)

[2] Duvall *et al.*, Phys. Rev. Applied **22**, 054030 (2024)

al “Chooz" approach. A selected
ed. Useful IBD events are those that
esult is taken from our previous

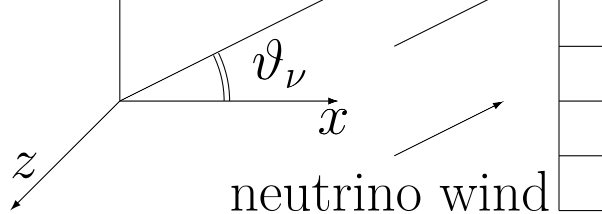
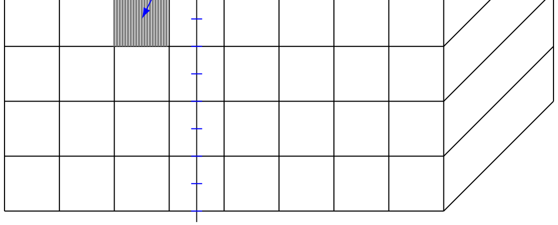
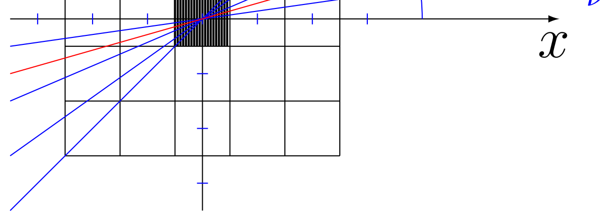


Figure 4: Diagram explai
orientation.

nd





plaining the prompt-delayed segment geometry, as well as neutrino and

Acknowledgments

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Administration award number DE-NA

and axes

10^{-} 10^{-}

Figure 7: Angular uncertainty plot comparing the pe
50 mm and 150 mm.

the Consortium for Monitoring, Technology, and Verification under Department of
-NA0003920. Prepared by LLNL under Contract DE-AC52-07NA27344. LLNL-POS

Counts n

e performance of three different square segment sizes: 5 mm,

t of Energy National Nuclear Security
-POST-XXXXXX.

Ref. [1]: